Target triage with protein language models: tractability priors and variant effects

Disease: ovarian cancer

Methods summary

All 15 of my top Module 2 targets mapped by gene symbol to reviewed human proteins in proteome.tsv (unmapped genes: none). To prevent leakage, I removed the 4 targets that were also in the training panel (COL1A2, COL3A1, MYH9 and TP53) before fitting any model, leaving 1,197 training proteins. Tractability priors are scaled logistic regressions on ESM C 300M embeddings (960 dimensions), trained on Open Targets clinical-precedence labels (Approved, Advanced Clinical or Phase 1; the "Druggable Family" label was excluded). I fitted separate priors for small molecules (sm_clinical) and antibodies (ab_clinical); the single labels used for validation and nearest-neighbour search are any_clinical. Variant scores come from a different model (ESM-2 650M) as log P(mutant) − log P(wild-type). Variant numbering follows UniProt precursor numbering throughout, so APOE R176C and C130R correspond to R158C (ε2) and C112R (ε4) in the mature protein. Data pack: UniProt release 2026_03.

1. Re-ranking

I combined the Week 2 score and the tractability prior as final = w × (percentile rank of Week 2 score) + (1 − w) × (percentile rank of max(p_sm, p_ab)). With w = 0.6, the top three targets stayed the same as a set (COL6A3, FN1, MUC16) but were reordered, with MUC16 falling from first to third. Three of the top five were retained (CHD4 and MXRA5 entered; COL3A1 and MACF1 left). The Week 2 score and the tractability prior alone were moderately correlated (Spearman ρ = 0.49, p = 0.064, n = 15), a suggestive but not statistically firm agreement given the small sample. The two lines of evidence are not expected to match closely, because one measures disease relevance and the other resemblance to past drug targets.

The largest shifts were three ranks: TP53 and MACF1 fell, whereas NOTCH2 and COL1A2 rose. TP53 fell because both modality priors were low (p_sm = 0.03, p_ab = 0.01), while NOTCH2 benefited from its antibody prior (p_ab = 0.20). However, NOTCH2 and MACF1 exceed the model input length, and small probability differences among mid-ranked targets make these shifts fragile.

Choice of w. I chose w = 0.6 because Week 2 evidence is disease-specific, whereas the tractability prior is only a modest predictor (family-split AUROC 0.741, AP 0.548). Sensitivity analysis (Table 1) showed stable extremes but some weight-dependent middle ranks, particularly NOTCH2.

Gene	Week 2 rank	Final, w=0.4	Final, w=0.6	Final, w=0.8	p_sm	p_ab
COL6A3	2	1	1	1	0.83	0.21
FN1	3	2	2	3	0.50	0.29
MUC16	1	4	3	2	0.00	0.15
CHD4	6	3	4	4	0.68	0.00
MXRA5	7	5	5	7	0.01	0.28
COL3A1	4	6	6	5	0.13	0.00
MACF1	4	10	7	6	0.07	0.00
COL1A2	11	8	8	11	0.19	0.00
KDM5C	8	11	9	8	0.09	0.00
PRKDC	10	9	9	9	0.14	0.00
NOTCH2	12	6	9	12	0.07	0.20
TP53	9	13	12	10	0.03	0.01
BPTF	14	12	13	14	0.14	0.00
MYH9	13	14	14	13	0.06	0.00
AHNAK	15	15	15	15	0.00	0.00

Table 1. Week 2 rank, final rank at three weights (w = 0.6 is the exported ranking), modality priors, and whether the sequence exceeds the ESM C input length. Tied ranks share a number.

2. Modality: where p_sm and p_ab differ

The sharpest split is CHD4 (p_sm = 0.68, p_ab = 0.00). CHD4 is a nuclear, ATP-dependent chromatin remodeler of the SNF2/helicase family and a core subunit of the NuRD complex. Because it is intracellular, it is out of reach of antibodies, which fits the near-zero antibody prior. Its ATPase domain offers a plausible ATP-binding pocket, consistent with the high small-molecule prior. Its nearest drugged proteins were PRKD1, PTK2B and CDK17 (cosine similarity 0.92 to 0.94), all kinases with ATP-binding pockets. CHD4 is an SNF2-family ATPase, not a kinase, so this resemblance does not

show that the same pocket chemistry applies. Its sequence (1,912 residues) fits within the model's input length, so this is the most trustworthy of the modality splits.

MXRA5 shows the opposite pattern (p_sm = 0.01, p_ab = 0.28): as a secreted matrix-associated protein without an annotated enzymatic domain, it is more naturally antibody-accessible. However, its modest p_ab and excessive sequence length (2,828 aa) make this tentative. Similar caution applies to COL6A3, whose high p_sm may reflect extracellular multi-domain architecture rather than a true small-molecule pocket. More generally, historical labels can undervalue novel modalities.

3. Validation

Split	AUROC	Average precision	AP chance level
Random 5-fold	0.840	0.717	0.332
Family-aware 5-fold (GroupKFold)	0.741	0.548	0.332

Table 2. Out-of-fold performance of the any_clinical prior on the 1,197-protein training panel.

The family-split AUROC of 0.741 and AP of 0.548 (chance 0.332) are the honest numbers. Whole families were held out, which matches how the prior is used here: scoring proteins that have no close drugged relatives in training. A random split puts paralogs in both training and test folds, so the model partly recognizes families instead of learning transferable features. This overstates performance by about 0.10 AUROC and 0.17 AP. Because the panel is enriched for drugged proteins (33% positive), I interpret outputs as rankings, not absolute probabilities.

4. Anchor variants

Gene	Variant	Note	Score	Percentile in protein
APOE	R176C	rs7412, defines ε2 (R158C mature)	−8.56	11.8
APOE	C130R	rs429358, defines ε4 (C112R mature)	+7.21	100.0
TREM2	R47H	rs75932628, AD risk variant	−5.16	47.7
TP53	R175H	hotspot	−5.97	19.3

Table 3. Anchor variant scores (ESM-2 650M; UniProt numbering). Higher percentile = more favoured substitution within that protein.

A strongly negative score indicates evolutionary constraint, not pathogenicity. APOE ε2 (R176C) is strongly negative (11.8th percentile) despite being protective, whereas risk-associated ε4 (C130R) is the most favored APOE substitution (100th percentile), plausibly because Arg is common at this position in other mammals. TREM2 R47H lies near the median (47.7th percentile), while TP53 R175H falls in the damaging tail (19.3rd percentile), showing that disease relevance does not necessarily track evolutionary constraint. I therefore compare percentiles within proteins rather than raw scores across proteins.

Sequence-only missense scoring also cannot capture splicing or regulatory effects, dosage, molecular interactions, tissue context, or mutant-specific mechanisms.

5. Limitations and next step

Sequence embeddings cannot establish 3D pocket geometry, accessibility, disease-context expression or causal effects of target modulation. Historical labels may also undervalue novel pockets or modalities. More importantly, nine of 15 targets exceed the ESM C input limit of 2,046 residues, and the data pack does not document how these sequences were handled. Priors and neighbours for these proteins may therefore represent only partial sequences, making conclusions for long targets such as COL6A3, FN1 and NOTCH2 less reliable than those for fully represented proteins such as CHD4 and TP53.

Targets for Week 4.

I would send CHD4 and PRKDC for structure-based pocket assessment.

- CHD4 (final rank 4): High p_sm, ATP-using neighbours, and a complete embedding make its ATPase pocket the strongest direct test of the sequence-based prior.
- PRKDC (final rank 9): Despite low p_sm (0.14) and an unreliable embedding due to its 4,128-aa length, its known PIKK-family ATP site can be assessed directly by structure.

Unmapped genes: None; All 15 genes mapped to reviewed UniProt proteins.